AZIZ MESSAOUD

Data Science Engineering Student | Applied AI/ML Engineer
Seeking 6-Month PFE Internship | Available January 2027 | Remote / Hybrid / On-site
+216 90 410 222 | Email | LinkedIn | GitHub | Portfolio

PROFILE

Final-year Data Science Engineering student at ESPRIT (2027) with production AI/ML experience in document intelligence, ranking systems, and agentic RAG. Seeking a 6-month PFE to contribute to agentic AI and knowledge management systems, with hands-on Python, FastAPI, LLM, and RAG pipeline skills.

TECHNICAL SKILLS

AI / ML: Python, scikit-learn, PyTorch, TensorFlow, NLP, LLMs, RAG, LangChain, Agentic AI, ranking, multi-agent systems, MLOps
Data & Engineering: Pandas, DuckDB, SQL, PySpark, FastAPI, Docker, Qdrant, REST APIs, Git, Linux

EXPERIENCE

AI Intern | Sopra HR Software — Jul–Sep 2026
Tunis, Tunisia
- Designed a seven-stage HR document-validation workflow (payroll, contracts, records) with explained flags and human review.
- Built a Docling, RapidOCR, and FastAPI pipeline to extract structured data from scanned HR documents end-to-end.

Machine Learning Engineering Intern | FlyRank AI — Jul–Sep 2026
Remote
- Built 32 features from 30K+ search-performance records for page-refresh ranking; reached 0.94 Precision@50 vs 0.76 fixed baseline on 1,583 held-out pages.

AI Intern | Sopra HR Software — Jul–Aug 2025
Tunis, Tunisia
- Automated entity extraction from unstructured HR PDFs with spaCy and pdfplumber, replacing manual extraction across standardized templates.

IT Support Intern | Banque de Tunisie — Jun–Jul 2023
Tunisia
- Supported helpdesk and IT operations in a regulated banking environment.

SELECTED PROJECTS

Tunisian Administrative Procedures RAG Assistant | BGE-M3, BM25, RRF, Qdrant, Mistral 7B, FastAPI, n8n
- **P:** Citizens cannot navigate dense multilingual (FR/AR/Darija) official documents without expert help.
- **S:** Hybrid RAG (BGE-M3 dense + BM25 fused via RRF), citation enforcement, answerability gating, Telegram delivery; 14 official sources, 286 chunks.
- **KPI:** Hit@5 0.873 vs 0.732 baseline (+19%) on 71 labelled questions.

TraffiX: Multi-Agent Traffic Management | Python, SUMO/TraCI, FastAPI, LLM APIs | Live | GitHub
- **P:** Urban traffic signals cannot coordinate dynamically under incident conditions, causing cascading delays.
- **S:** Led control subsystem (A1 signal authority, deterministic safety mask, recovery logic) for a 6-junction Tunis SUMO simulation in a 6-person team.
- **KPI:** Peak pedestrian wait 116.3s vs 144.1s baseline on real Tunis OSM road topology.

FlyRank Content Refresh Ranker | Python, scikit-learn, DuckDB, RandomForest
- **P:** Content teams have no data-driven signal to prioritize pages for refresh under a limited review budget.
- **S:** 32 features engineered from 30K+ search-performance records; Random Forest with client-holdout split on FlyRank's production data warehouse.
- **KPI:** Precision@50 0.76 to 0.94 on 1,583 held-out pages (24% lift over rule-based baseline).

EDUCATION

Engineering Degree in Data Science | ESPRIT, Tunis | Expected 2027
Technological Baccalaureate | Lycée Mahmoud Messaoudi, Nabeul | 2021

CERTIFICATIONS

OCI Data Science Professional (2025) · OCI AI Foundations (2025) · Azure AI-900 (2026) · Azure DP-900 (2026) · Databricks Generative AI (2026)
NVIDIA Fundamentals of Deep Learning (2025) · Getting Started with Deep Learning (2025) · Transformer-Based NLP (2025) · Samsung Innovation Campus

LEADERSHIP & COMMUNITY

Happy City Index Researcher (2026, Tunis data) · **DeepFlow AI** (RAG/vector-search workshops) · **ATIA Club ESB** (AI workshops, Learning Dept.) · IEEE member (2022–2023) · Theirworld Global Youth Ambassador